

DHILLON PATEL

San Francisco, CA | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | dhillonpatel21@gmail.com | 260-450-5921

SKILLS

Languages & Tools: Python, C, C++, Git, Jenkins, Linux, Docker, Yocto

Networking: VPN Translation, ECMP, BIRD Internet Routing Daemon, BGP, routing protocols

AI/ML: Multi-agent LLM workflows, graph-based orchestration, PyTorch, agentic tooling for CI/CD

WORK EXPERIENCE

Cisco Meraki

San Francisco, CA

Software Engineer, MX Routing Team

March 2025 – Present

- Leading the team's AI initiative, architecting a multi-agent LLM workflow that takes a ticket end-to-end to a merge request, using a graph-based engineering approach to orchestrate planning, code generation, and review agents
- Design and maintain VPN NAT logic supporting 1:many and many:many address mappings for MX Security Appliances, conserving IP address space across large-scale customer deployments
- Develop and extend ECMP (Equal-Cost Multi-Path) routing and BIRD-based dynamic routing to improve traffic distribution, failover behavior, and route table reliability
- Overhauled the route table page by introducing protobuf-based firmware-side filtering, drastically reducing page load times for large route tables
- Built a Linux ip-style wrapper script for commonly forgotten table read handlers, cutting copy/paste errors and producing cleaner output — saving roughly 10 seconds per lookup for the ~500-person support team, who run these dozens of times a day

Nordson

Amherst, OH

Embedded Software Engineer

October 2024 – February 2025

- Developed embedded software for STM32 microcontrollers, configuring JTAG, pin configurations, and integrating an Ethernet PHY and Ethernet switch chip
- Designed a command-line interface for the bootloader, reducing software download time by 20 seconds compared to the GUI version

Cisco Meraki

San Francisco, CA

Firmware Software Engineer Intern

September 2023 – December 2023

- Designed and integrated new testing pipelines using Jenkins and Gerrit, accelerating the team's development process
- Added code coverage to native test recipes, improving test effectiveness by 30% through CMake, Makefiles, Autotools, Ruby scripts, and BitBake recipes in Yocto

PROJECTS

RGB-D Semantic Segmentation

- Trained multiple semantic segmentation models in Python with PyTorch, demonstrating improved accuracy on the NYUv2 dataset after adding a depth channel; presented research paper to faculty

Autonomous Marco Polo Robots

- Implemented pursuit-evasion behaviors for two differential-wheeled robots in Python and C++, integrating LCM and OpenCV for real-time audio and visual tracking on Raspberry Pi 4 with an April Tag fiducial system

Concussion Detector

- Researched and built a wearable IMU-based neck/head sensor with a touchscreen watch interface for real-time concussion/whiplash prediction, using STM32 microcontrollers, UART, I2C, SPI, and C drivers

EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Computer Engineering | GPA: 3.87

August 2020 – May 2024

Relevant Courses: Computer Vision, Embedded Control Systems, Autonomous Robotics Design Experience

Awards: Summa Cum Laude, James B. Angell Scholar, Constance B. Mathias Engineering Scholarship, Richard Earhart Scholarship, Dean's Honor List, University Honors, William J. Branstrom Freshman Prize